

1 Introduction

The purpose of this short course is to cover a range of principles and actual devices used to detect photons and ionized particles. The variety of the detectable particles is very large, and the scope of the course must restrict it in some ways.

Radiation Detectors

A fundamental feature of nuclear processes is that the energy released is larger than the binding energies of atomic electrons. Any emitted particles will have sufficient energy to ionize atoms. Nuclear radiation is called "ionizing radiation", therefore, and detecting this ionization allows us to observe nuclear processes. Radiations that interact with matter via the electromagnetic force, *i.e.*, electrons, charged particles and photons, can directly ionize or excite atoms. These radiations are readily detected. Neutrons interact with nuclei only via the nuclear force and are detected through indirect or secondary ionization processes.

Radiation can be described as energy or particles from a source that travel through space or other mediums. Light, heat, and the microwaves and radio waves used for wireless communications are all forms of radiation.

Radiation includes particles and electromagnetic waves that are emitted by some materials and carry energy. The kind of radiation discussed below is called **ionising radiation** because it can produce charged particles (or **ions**) in matter. X-rays, gamma-rays, alpha particles, beta particles and neutrons are all examples of ionising radiation

Types of Radiation:

- a) charged particles
electrons, protons, atomic nuclei
+ many elementary particles
- b) neutral particles
neutrons
+ many elementary particles
- c) photons
light
x-rays
gamma rays

- A photon is a "packet" of electromagnetic energy.

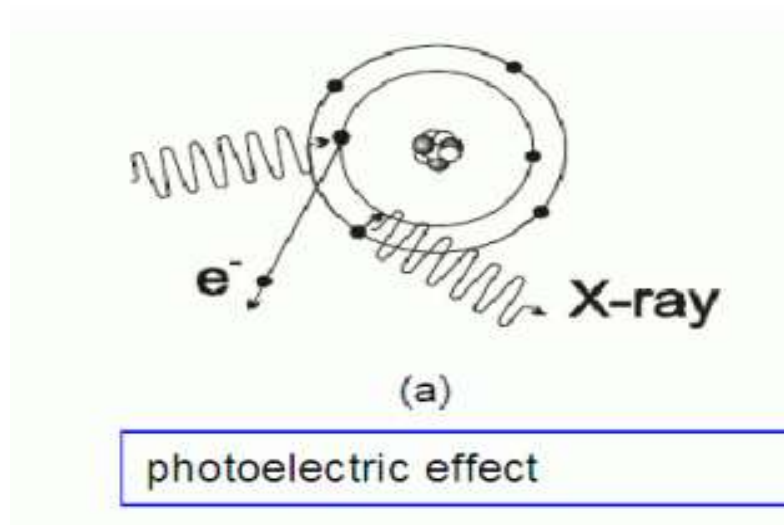
It has no mass, no charge, and travels in a straight line at the speed of light.

- Many instruments have been developed to detect radiation
- Based on knowledge of how radiation interacts with matter
 - Excitation
 - Ionization
- Charged particles cause ionization directly through Coulombic interactions
- EM radiation produces ion pairs in matter
 - Photoelectric effect
 - Compton scattering
 - Pair production
- Neutrons produce ions through secondary mechanisms

In the energy range from visible light to g-rays, then following processes are the main forms of interaction between the photons and matter:

2.1.1 Photoelectric effect

A photon interacts with a bound electron in an atomic or molecular orbital or in an energy band of a solid (Figure 2a):



The photoelectric effect thus creates free electrons that can be collected using electric fields and handled by electronic circuits. It is the basis of many radiation detectors and ought to be studied more closely. In atomic photoemission the departing photoelectron obtains the kinetic energy

$$E_k = h\nu - E_b(nl), \quad (1-2)$$

where $E_b(nl)$ is the binding energy of electrons in the atomic orbital with the quantum numbers nl . The lowest energy of the radiation that can cause photoemission in atoms is thus

$$h\nu_{\min} = I(A), \quad (1-3)$$

where I is the **ionization energy** (the binding energy of the outermost electron shell) of the element A. As seen from Figure 3, it varies greatly, being smallest for the alkali metals and highest for rare gas atoms.

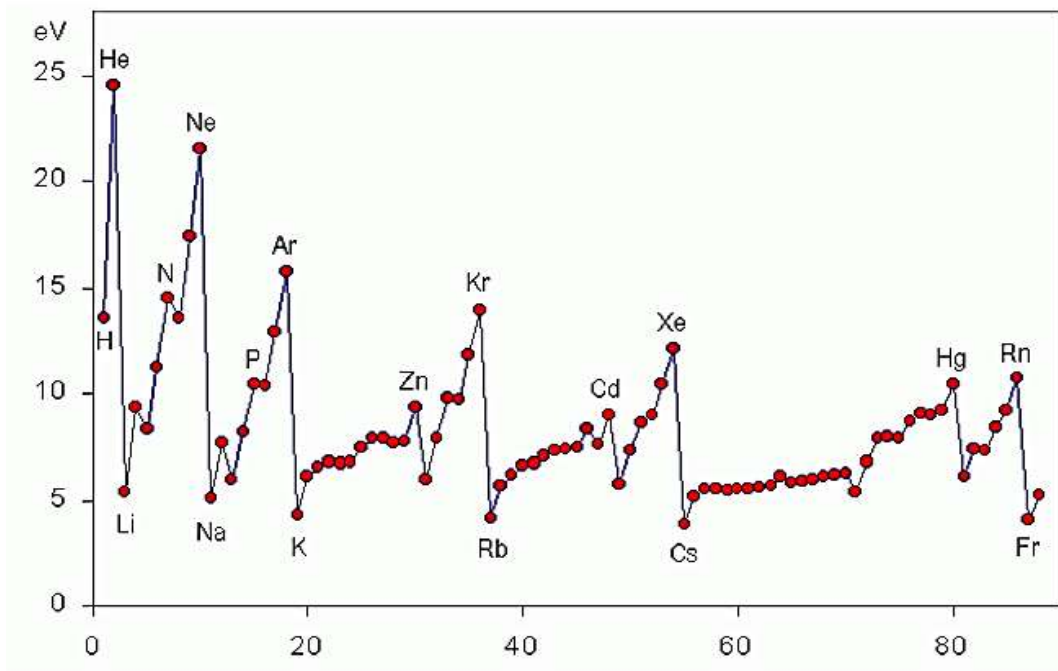


Figure 3. Ionization energies of free atoms.

In solids, the electrons form electron bands due to the interaction between the discrete energy levels of many individual atoms. The bands can be regarded as distributions of very closely packed quantum states. Electrons are fermions and only one electron can occupy each quantum state of the system. The overall band structure is characterized by the *density of states* function $D(E)$, which tells how many electrons can be in an energy range $E \dots E+dE$. The quantum states of the bands are filled up to the **Fermi level** E_F . The electrical properties of the material are determined by the position of the Fermi level (Figure 4). In semiconductors or insulators, the highest band containing

electrons is referred to as the **valence band** and the next empty band above it the **conduction band**. In metals, this distinction is not clear and the electrons in the uppermost band are named either *valence* or *conduction* electrons in different sources. In semiconductors, the band gap is relatively small and the energy separation between the valence and conduction bands in semiconductors can be overcome by thermal, electrical, or chemical excitations.

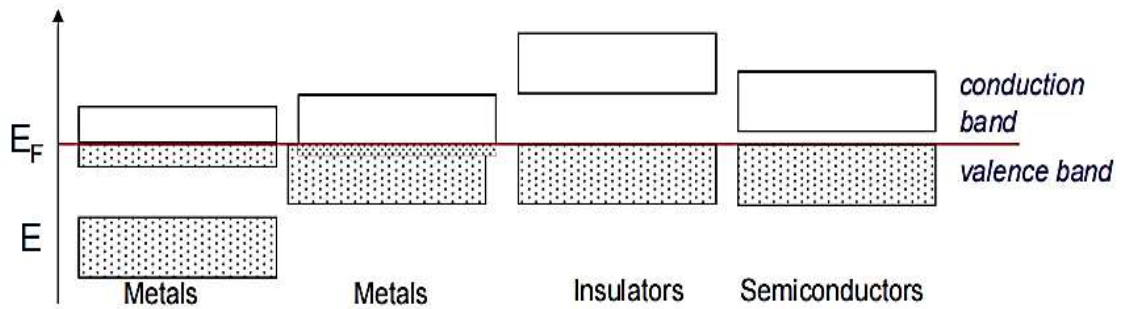


Figure 4. Classification of solids according to their band structure.

In detectors, the most relevant is the photoemission from metal surfaces or photoabsorption within semiconducting materials, as illustrated in Figure 5. The electron binding energy in metals is measured from the position Fermi level and one needs to take into account that additional energy – **work function** Φ – is required to completely remove the photoelectron from the material into vacuum, so that

$$E_{kin} = h\nu - E_b - \Phi. \quad (1-4)$$

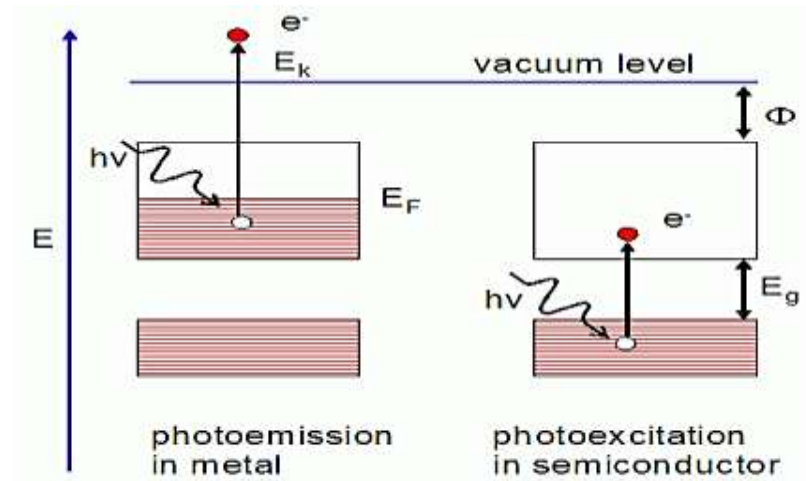


Figure 5. Two photoabsorption processes relevant to particle detection.

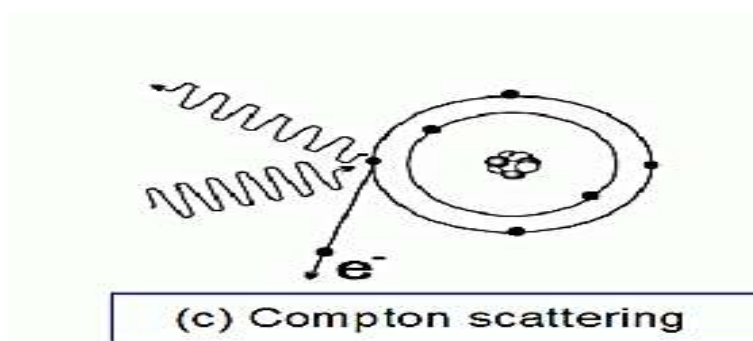
In semiconductor-based detectors, we are not interested in removing the charge carriers completely from the material, but in lifting a valence-band electron to the conduction band and creating electron-hole pairs. These electrons and holes can then drift through the material and be collected on electrodes. The minimum photon energy for creation such an electron-hole pair is

$$h\nu_{\min} = E_g, \quad (1-6)$$

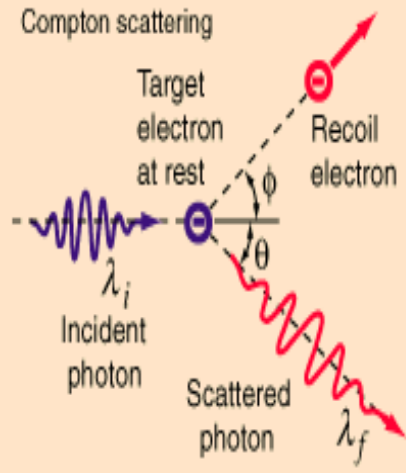
where E_g is the band gap of the material.

Compton Effect

Compton effect or **Compton scatter** is one of principle forms of photon interaction. It is the main cause of scattered radiation in a material. It occurs due to the interaction of the photon (x-ray or gamma) with free electrons (unattached to atoms) or loosely bound valence shell (outer shell) electrons. The resultant incident photon is scattered (changes direction) and imparts energy to the electron (recoil electron). The scattered photon will have a different wavelength (observed phenomenon) and thus a different energy ($E=hc/\lambda$). Energy and momentum are conserved in this process.



Compton Scattering



Compton scattering

Target electron at rest

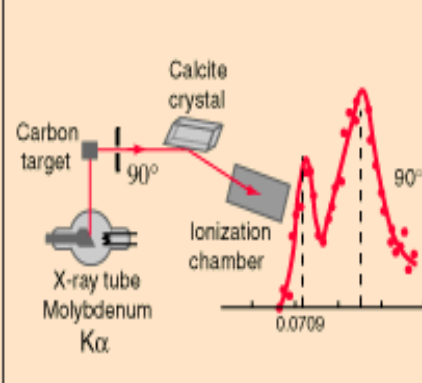
Recoil electron

Incident photon λ_i

Scattered photon λ_f

Arthur H. Compton observed the scattering of [x-rays](#) from electrons in a carbon target and found scattered x-rays with a longer wavelength than those incident upon the target. The shift of the wavelength increased with scattering angle according to the Compton formula:

The Compton effect is a partial absorption process and as the original photon has lost energy, known as Compton shift (i.e. a shift of wavelength/frequency). The wavelength change of the scattered photon can be determined by $0.024 (1 - \cos \theta)$, where θ is scattered photon angle. Thus, the energy of the scattered photon decreases with increasing scattered photon angle. The higher incident photon energy will cause a greater percentage loss of their energy during Compton scatter while lower energy photons will lose a smaller percentage of their energy.

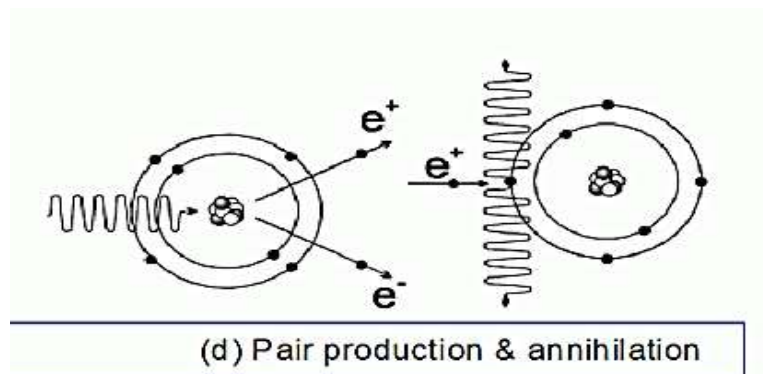


$$\lambda_f - \lambda_i = \Delta\lambda = \frac{h}{m_e c} (1 - \cos \theta)$$

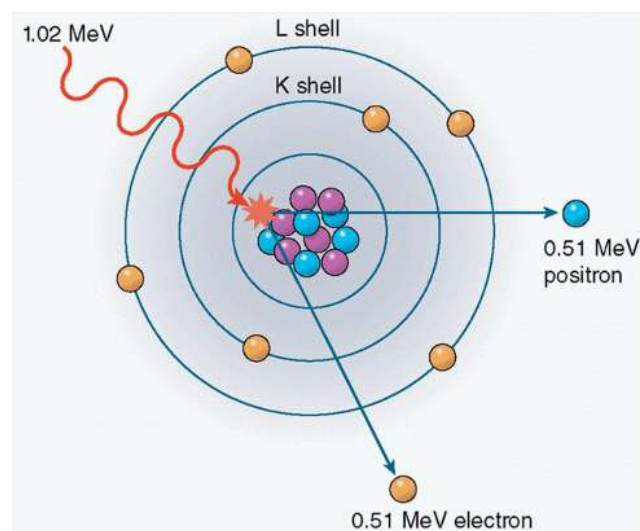
Compton explained and modeled the data by assuming a particle (photon) nature for light and applying conservation of energy and conservation of momentum to the collision between the photon and the electron. The scattered photon has lower energy and therefore a longer wavelength according to the [Planck relationship](#).

Click for more detail

Pair production



Pair production is not significant in diagnostic X-ray imaging and requires extremely high energies for it to occur. Pair production can only occur when the energies of X-rays exceed 1.02 MeV. In pair production, an X-ray photon interacts with the electric field of the nucleus of an atom and the photon energy is transformed into an electron-positron pair. Each electron has a rest mass energy equivalent of 0.511 MeV, hence the energy threshold for this reaction to occur is 1.02 MeV. When photon energy is greater than the threshold, the excess is transferred to the electrons as kinetic energy. In turn, the electron and positron lose their kinetic energy via excitation and ionization. A positron may interact with a negatively charged electron as it comes to rest, resulting in the formation of two oppositely directed 0.511 MeV annihilation photons (Figure 2.6).



☞ The minimum energy required is

$$E_{\gamma} \geq 2m_e c^2 + \frac{2m_e^2 c^2}{m_{\text{nucleus}}} \approx 2m_e c^2 = 1.022 \text{ MeV}$$

☞ The process is more probable with a **heavy nucleus** and incident **gamma rays with higher energies**.

Probability of Interactions

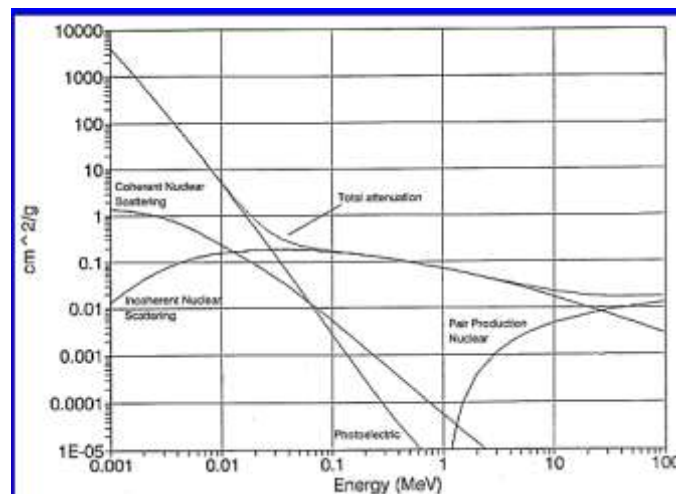
- Charged particles (e.g., alphas and betas) continuously interact with the material they travel through. Photons occasionally interact.
- We will now consider the factors that affect the probability of photon interactions (photoelectric effect, Compton scattering, pair production)

The two most important factors influencing the probability of a given type of interaction are: Probability of Interactions

- The energy of the photon (E) - the atomic number (Z) and density of the material (i.e., the electron density)

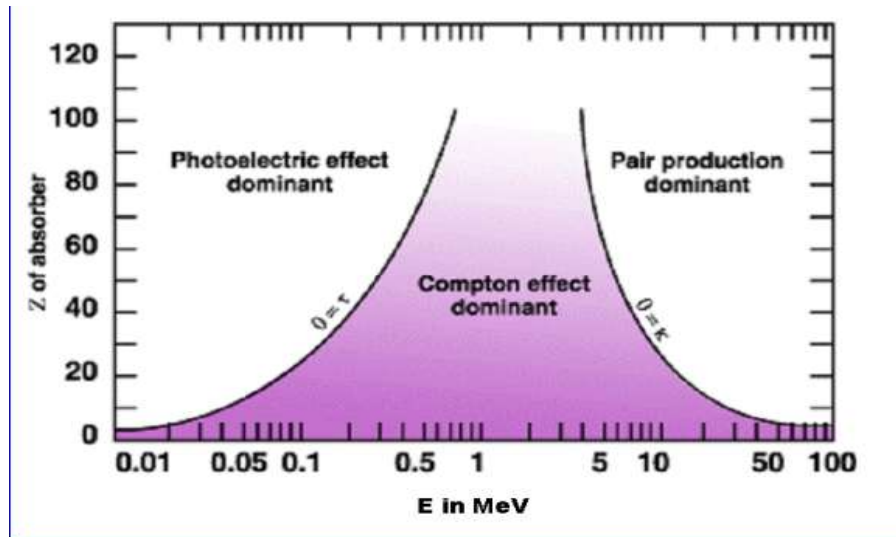
The next slide shows the probability of an interaction as a function of the photon energy in a low-Z material (water).

- Note that the probability of the photoelectric effect decreases dramatically as the photon energy increases.



The slide below shows how the photoelectric effect is dominant at low energies in high Z materials.

- Compton scattering occurs at all energies in all materials, but is the dominant interaction in the medium energy range.
- Pair production only occurs at energies above 1.022 MeV. Pair production becomes more probable as the photon energy and the atomic number of the material increase.



There are two common quantities used to describe the probabilities of these interactions: linear attenuation coefficient (μ), mass attenuation coefficient (μ / ρ).

The linear attenuation coefficient (μ) is the probability of an interaction per linear distance traveled by the photon.

- It is actually the sum of the probabilities of the photoelectric effect (τ), Compton scattering (σ) and pair production (κ)

$$\mu = \tau + \sigma + \kappa$$